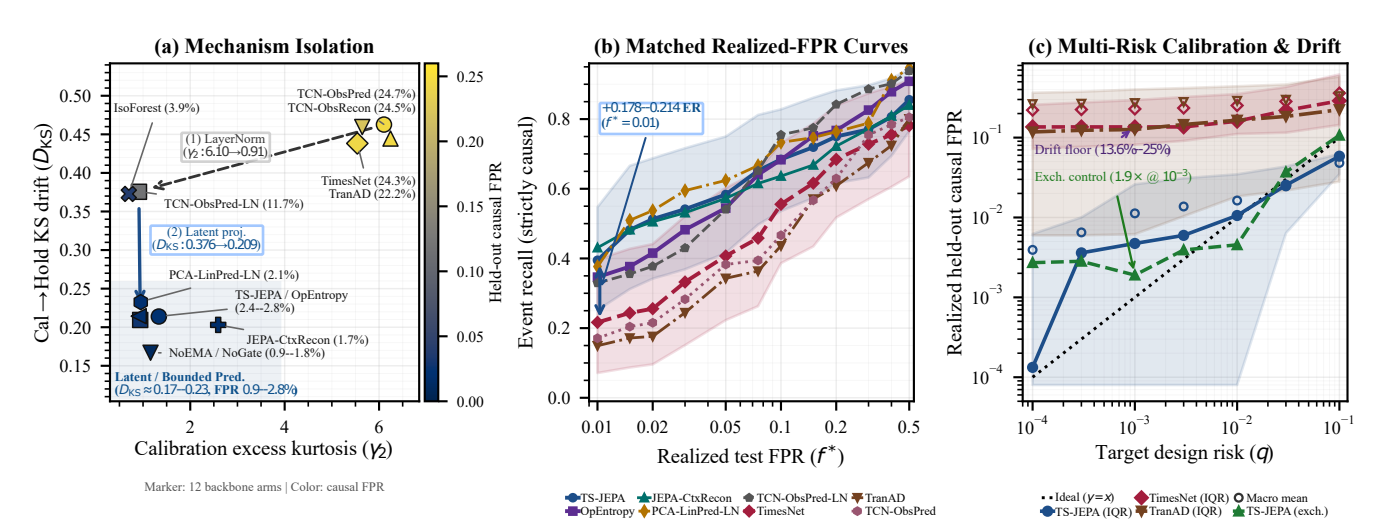

(a) Mechanism Isolation
Cal→Hold KS drift ($D_{KS}$)
Calibration excess kurtosis ($\gamma_2$)
Held-out causal FPR
IsoForest (3.9%)
TCN-ObsPred (24.7%)
TCN-ObsRecon (24.5%)
(1) LayerNorm ($\gamma_2$: 6.10→0.91)
TimesNet (24.3%)
TranAD (22.2%)
TCN-ObsPred-LN (11.7%)
(2) Latent proj. ($D_{KS}$: 0.376→0.209)
PCA-LinPred-LN (2.1%)
TS-JEPA / OpEntropy (2.4–2.8%)
JEPA-CtxRecon (1.7%)
NoEMA / NoGate (0.9–1.8%)
Latent / Bounded Pred. ($D_{KS}$ ≈ 0.17–0.23, FPR 0.9–2.8%)
(b) Matched Realized-FPR Curves
Event recall (strictly causal)
Realized test FPR ($f^*$)
+0.178–0.214 ER ($f^*$ = 0.01)
TS-JEPA
JEPA-CtxRecon
TCN-ObsPred-LN
TranAD
OpEntropy
PCA-LinPred-LN
TimesNet
TCN-ObsPred
(c) Multi-Risk Calibration & Drift
Realized held-out causal FPR
Target design risk ($q$)
Drift floor (13.6%–25%)
Exch. control (1.9× @ $10^{-3}$)
Ideal ($y=x$)
TimesNet (IQR)
Macro mean
TS-JEPA (IQR)
TranAD (IQR)
TS-JEPA (exch.)
Marker: 12 backbone arms | Color: causal FPR